

EC-417: Introduction to Machine Intelligence

Assignment 1 — Bias–Variance Tradeoff and Exploratory Data Analysis

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Reproducibility: Python 3.11.7. Exact package versions are recorded in requirements.txt. Random seeds were fixed in the experiments.

AI disclosure: AI assistance was used for planning, debugging, code review, explanation of machine-learning concepts, and report organization. Code was executed locally and numerical outputs and figures were checked by the student.

1. Environment, Version Control and Vectorization

Two pairwise Euclidean-distance implementations were evaluated: explicit nested Python loops and NumPy broadcasting. An additional norm-expansion implementation was also evaluated. The correctness checks passed with the required tolerance. For $N=2000$ and $d=20$, broadcasting was substantially faster than explicit loops, while norm expansion was faster still and avoided the $O(N^2d)$ intermediate array used by direct broadcasting.

Method	Time (s)	Speedup
Nested loops	35.358	1.00×
Broadcasting	0.777	$\approx 42.93\times$
Norm expansion	0.0227	$\approx 1472\times$

Direct broadcasting requires an intermediate array of shape $N \times N \times d$, so memory grows as $O(N^2d)$. Norm expansion reduces the main intermediate to $O(N^2)$. For large datasets, norm expansion is attractive when numerical behavior is controlled; chunking or other memory-aware approaches may be preferable when the full $N \times N$ distance matrix itself is too large.

2. Polynomial Curve Fitting

The data-generating process was $y = \sin(2\pi x) + \varepsilon$ with $\sigma=0.3$, x uniformly distributed on $[0,1]$, and fixed random seeds. Polynomial degrees 0 through 12 were evaluated for $n=15$ and $n=100$ training observations.

$n=15$: The minimum test RMSE occurred at $M=3$, while training RMSE continued to decrease and was smallest at $M=12$. The largest generalization gap occurred at $M=12$. This demonstrates that minimizing training error alone does not select the best predictive complexity.

$n=100$: The test-error curve became substantially more stable and the severe high-degree overfitting seen with $n=15$ was reduced. Increasing the training size lowers variance, making flexible models less unstable.

Bias/variance interpretation: Increasing polynomial degree generally reduces bias but increases variance. Increasing the training-set size mainly reduces variance; the irreducible noise level remains fixed.

Degree 9 with $n=15$: A very small training RMSE is not evidence of good prediction because a high-degree polynomial can closely fit noisy training observations while behaving very poorly on independent inputs.

$n=15$ result	Value
Minimum test RMSE degree	$M=3$
Minimum training RMSE degree	$M=12$
Largest generalization gap	$M=12$

3. Empirical Verification of the Bias–Variance Decomposition

For degrees $M \in \{1, 3, 9\}$, 200 independent training sets of size 15 were simulated on a 100-point grid. The average prediction, pointwise bias², variance, and simulated test loss were computed. A fresh noise sample was used for the simulated total, as required by the assignment.

M	Noise σ^2	Bias ²	Variance	Sum	S=2000 Total	Discrepancy
1	0.090000	0.205602	0.047467	0.343269	0.343472	0.000203
3	0.090000	0.005599	0.058439	0.154038	0.154482	0.000444
9	0.090000	2235.660139	17598065.184776	17599300.934915	17593104.320627	3.385713

The largest absolute discrepancy in the S=2000 calculation was approximately 3.386 for M=9. Although this looks large in isolation, the degree-9 total error is itself extremely large because the polynomial is numerically unstable and has very high variance. The decomposition is an expectation identity, while the experiment uses finite Monte Carlo samples, so a nonzero discrepancy is expected. Increasing S from 200 to 2000 provides a more accurate approximation.

Bias–variance trend: Low-degree models show larger bias and relatively low variance; high-degree models show much larger variance. The exact decomposition cannot be computed on a real dataset because the true function f^* and the noise variance σ^2 are not known. Consequently, bias–variance decomposition is best used as a conceptual and empirical framework rather than as an exactly observable quantity for real-world data.

4. Exploratory Data Analysis — California Housing

The California Housing dataset contains 20,640 census block groups with eight numerical predictors and one continuous target. The analysis checked data types, missing values, summary statistics, distributions, correlations, geographic structure, and preprocessing leakage.

Target: The exact maximum target value is 5.0. The sharp upper cutoff is right-censoring (top-coding). Expensive districts above the recording ceiling are therefore represented by the same target value, causing a regression model trained on these labels to systematically underestimate the most expensive districts.

Predictor quality: AveRooms, AveBedrms and AveOccup exhibit severe right skew and unusually large values. Because these are ratio variables, extreme ratios can occur when denominators are small. They should be investigated and treated using an appropriate transformation or robust method rather than automatically deleted.

Correlation and geography: MedInc has the strongest direct association with the target. Latitude and Longitude provide important geographic information. The geographic scatter plot shows high-value concentrations around the Bay Area and Southern California. Nearby blocks can share unobserved local factors, so a random split can place highly related observations in both train and test sets.

Data leakage: Standardization must be fitted on the training split only. Fitting the scaler on the entire dataset lets test-set information influence preprocessing. Under strong distribution shift—for example, a future-time or geographically different test set—the difference between the two procedures can become much larger.

5. Plain-Language Summary of Problem 4

The California Housing dataset describes 20,640 geographic areas in California using information about income, housing, population and location. The goal of the exploratory analysis was to understand the data before building a model.

The first major data-quality problem is that the recorded house-value target stops at 5.0. This is called top-coding or right-censoring. It means that very expensive areas above the recording limit are all given the same value. A model trained on these data can therefore underestimate the values of the most expensive districts.

The second major problem is the presence of extreme values in ratio-based variables, especially average rooms, average bedrooms and average occupancy. Very small denominators can create unusually large ratios. These observations should be investigated rather than automatically deleted.

The three predictors to examine first are median income, latitude and longitude. Median income provides a strong economic signal associated with house value. Latitude and longitude capture strong geographic differences in housing prices. The geographic analysis also shows that nearby areas can be similar in ways not directly measured by the dataset.

Finally, preprocessing must be learned from the training data only. If the complete dataset is used to calculate scaling statistics before the train/test split, information from the test data leaks into the analysis. This can make evaluation less representative of performance on genuinely unseen data.

Generated figures: The computational scripts save the required figures programmatically under the repository's *figures/* directory. These include the Problem 2 polynomial-fit and RMSE plots, Problem 3 bias–variance plots, and Problem 4 EDA plots.